Southern New Hampshire University

CS-360— Project 1, Assignment 3-3: Proposal for an Inventory Application for the Android mobile phone

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1 Introduction

This paper constitutes a project that I am tasked with working on regarding the development of an Android mobile phone application to demonstrate the principles of mobile architecture and software development. In particular, I will assume the role of a software developer for the fictious *Mobile2App* company.[1] While "working" for this firm, I do have some creative control— in particular with selecting what kind of mobile application that I can develop. I have decided to work with an "inventory app," which will be used to keep track of items in a warehouse.

1.1 Ergonomic Problems for the Industrial Worker

In the business world, both managers and workers are faced with problems in *ergonomics*, which Wilson (2000, p. 560) has defined as "the theoretical and fundamental understanding of human behaviour and performance in purposeful interacting sociotechnical systems, and the application of that understanding to design of interactions in the context of real settings." In particular, a subset of the business world is tasked with the mass-production and dissemination of a homogenous product, or an "inventory item," which is a problem that increases in difficulty as inventory items increase in both count and diversity.

A *warehouse* is a fairly large building where the inventory items, great in their diversity,[2] are stored for some elongated period of time, later to be sold as they are

[1] See CS-360 (n.d., §Scenario).

[2] An index to quantify "diversity" of different kinds, with an R total count of kinds and a frequency by each kind p_i, comes in the form of the entropy formula devised by C. Shannon, formally put as: $H' = -\sum_{i=1}^{R} p_i \log_e(p_i)$ —this may be useful in the application of measuring the different kinds of items and their respective inventories.

demanded by consumers. As a company grows, they may need more warehouses to store the different kinds of homogenous products. A simple accounting system consisting of a ledger and a pen, or pencil, by which to jot down transactions is typically used to track inventories, simply does not scale in regards to the capacity to store and process large quantities of information.[3] Modern internetworked computer and information systems hold much potential in the solution to these problems.

2 Proposal for an "Inventory App"

Computer and information systems have shown themselves to be helpful in the industrial world. Prior to the releaase of mobile phones[4] and portable computer systems,[5] the business world has invented a number of quantitative techniques for the efficient production, storage, and distribution of material goods.[6] The rubric (CS-360, n.d., §Option 1) lays out the requirements for what is needed for the inventory.[7]

Regarding the inventory app's *model*, I intend to use a standard relational database to store both inventory items and user credentials. Regarding the inventory app's *controller*, I intend to design and implement mechanisms for adding and removing items from the inventory table, mechanisms for keeping track of the count of the specific kinds of items in the inventory, and a mechanism by which

[3] I elaborate further on the limitations of the "analog method" in later sections.

[4] Circa 2007 with the Apple Company's release of the first iPhone.

[5] Circa 1984 with the Apple Company's release of the Macintosh 128K computer.

[6] See the following reference book for an application of the quantitative methods to business and economic problems: Downing, D., Clark, J., and Friedlob, G. T. (1988). *Quantitative Methods: A Streamlined Course For Students & Business People*. Barron's Business Review Series.

[7] Here, I address the directives CS-360 (n.d., Directions, §I.A and §I.C).

to notify the end-user when items reach a count for zero. Depending on how the inventory app's requirements evolve, I may need to introduce additional mechanisms to alert the end-user in case of over-supply or surplus of items, in addition to the given required mechanism for alerting the end-user for a zero count.[8]

Finally, regarding the inventory app's view component, I will need to, at the very least, build a screen for listing the items in the inventory. I think that I should apply a `TableLayout` in Android studio to show the tabular dataset embodied by the SQL database's listing of inventory items and their respective counts. Through the inventory app's controller, I will send notifications to the inventory app's view component, which will hopefully alert its end-user when zero counts of a kind of inventory items has been met.

2.1 Inventory App as a Solution to Ergonomic Problems

An inventory app mainly solves problems in regards to ergonomics. In particular, the inventory app is the "sociotechnical" solution to the problem of efficiently allocating resources by allowing for quicker storage and processing of information regarding a warehouse's inventory.[9]

While the relatively simple bit of machinery that consists of ledgers and pens to manually record and process the information exists, which I shall refer to as the "analog method," two limitations arise from this approach: first, the "analog method" does not scale as the size and number of warehouses are used to store and process information. Second, with multiple workers indirectly collaborating with one-another, the "analog method" could lead to confusion when adding, up-

[8] I think that this could be a good feature, but I will only implement if given authorization.

[9] This subsection addresses CS-360 (n.d., §I. B.) in the rubric.

dating, querying, and dropping information regarding the inventory. The modern approach of mobile phone applications solves these two problems by corrobrating the recorded work of many workers into a single information system, with near-perfect logging and auditing methods, and superior processing and storage of the data.

I claim that in the case of a diverse and large network of warehouses and their respective workers, the mobile phone application is superior to the "analog method" in terms of reducing errors, and efficient processing and storage of the inventory data. This solves the ergonomic problem of developing a useful component to the sociotechnical system for more efficiently organising workers into doing their inventory accounting and work, regarding the storage and processing of information about the items in the inventory, with minimal errors.

3 Discussion

3.1 Target Audience for the Inventory App

In particular, the inventory app's primary target audience are the warehouse managers and the workers that the managerial class is in charge of[10]— in particular so that both management and workers can keep track of inventory items as they come into, and out of, their respective inventories or the warehouse. In this case of these two kinds of users, different permissions need to be given to the managers and workers accounts. Managers need to have more direct access to the inventory database, whereas the labourers should have more specific permissions to the

[10]This adresses two of three possible kinds of "end-user" for the inventory app (CS-360, n.d., §III. A.).

inventory database.

As discussed earlier, internetworked computer systems are ideal for a vast network of warehouses and their respective workers, along with management who needs to organise the worker's labour power.[11] In the case of both kinds of end-users, they are likely to use the inventory app because of the advantages offered by mobile apps compared to the aforementioned "analog method" for storing and processing "meta-information" about inventory items.[12] With proper user-interface design and minimal training needed to use the inventory app, it will demonstrate itself to be useful in an efficient workflow for inventory management at large scales.[13] In the ideal, very little interaction will be with the inventory app in order to work with inventory items in its database.[14]

A third kind of end-user for this inventory app could be anyone who needs to keep track of some material or tangible thing in their personal lives.[15] I think that, in some special cases, a computer system can show itself to be useful for some individuals. Of course, the "analog method" that consists of a "to-do list" might be more helpful for individuals, especially if they are not "tech-savvy." But there may be a niche subset of individuals who prefer to use mobile phones for reminders or personal inventory management, so this class of end-users could find value in the inventory app.

[11]This "touches on" CS-360 (n.d., §III. B. i.).
[12]This addresses CS-360 (n.d., §III. B., ii.).
[13]This addresses CS-360 (n.d., §III. B., iii.).
[14]This addresses (CS-360, n.d., §III. B., iv.).
[15]This introduces a third class of end-users for the inventory app, fulfilling the target count of "kinds of end-users" as instructed by the rubric (CS-360, n.d., §III. A.).

	Crunchtime Inventory (Google Play, n.d.-a)	**Collection & Inventory Tracker (Google Play, n.d.-b)**
Listing Tasks	✓	
Recording Transactions	✓	
Place Orders	✓	
Categorizing Items		✓
Tracking by Barcode		✓
Automatic Sorting		✓
Dropdown Menu		✓

Table 1: Comparison between different mobile applications for inventory item management

3.2 Alternatives to the Inventory App

While doing a cursory research for this project and drawing inspiration from prior art,[16] I have come across two released mobile phone apps for inventory management: *Crunchtime Inventory* and the *Collection & Inventory Tracker*. They both solve the ergonomic problem of inventory item management.

Crunchtime Inventory has the features of listing tasks for workers, inventory count, a mechanism to record transactions of inventory items, and another mechanism to place orders with vendors. Collection & Inventory Tracker offers the features of categorizing inventory items, tracking said items by barcode, storing additional information about the item regarding specific fields and properties, a mechanism to update items in the inventory, and "dropdown items and tags." Table 1 depicts a quick summary of their features as derived from analysing their features from *Google Play*.

It is tempting to combine all of the features presented in the comparison table shown above, but such an approach might lead to feature creep.[17] To engineer an

[16] In accordance with the rubric (CS-360, n.d., §II. A.).

[17] "Feature Creep" is the introduction of too many features into a product, which may lead to

efficient and "lean" tool for inventory item management, further research will be needed to conducted on the target user base, and I will find the most parsimonious set of features to include to the "inventory app."[18]

4 Summary

The following are a condensed explaination of the kind of problem faced by factory workers, the proposed solution and its abridged specifications, and potential value added by the tool,

- A problem of not just efficient manufacturing of economic goods, but also the efficient storage and accounting of such economic goods, is of the upmost priority to solve.
- A standard accounting system consisting of a ledger and pen-and-pencil might be suited for smaller warehouses and for small networks, but as the warehouses grow in both individual size and in collective count, this "analog method" does not scale, and a transfer to digital and internetworked computer and information systems become more necessary.
- I propose a solution that exploits modern computing and information systems that comes in the form of a mobile phone application, which I shall call the "Inventory App," that allows for quicker processing and storing of the "meta-information" regarding the items and their respective inventories.

increased complexity and increased possibility of inefficies and failure; see: Sullivan, J. M. (n.d.). Impediments to and incentives for automation in the Air Force. *Proceedings. 2005 International Symposium on Technology and Society, 2005. Weapons and Wires: Prevention and Safety in a Time of Fear*. ISTAS 2005. https://doi.org/10.1109/istas.2005.1452719

[18] A "modular design" can be applied to the inventory app so that end-users can install plugins if they want additional features, but this will need to be thoroughly tested for effectiveness.

- Inventory app is primarily targeted to warehouse managers and workers, but they can be repurposed by the non-manager or non-worker where problems of inventory accounting is needed, though it should be noted that a computer and management system for individual users may be cumbersome, and may only apply to special cases. Nonetheless, the inventory app needs to take all three classes of users into account, but put more priority to the warehouse managers and workers.

- To provide a baseline of comparison, I have cited two prior arts, *Crunchtime Inventory* (Google Play, n.d.-a) and *Collection & Inventory Tracker* (Google Play, n.d.-b), by which to use as a basis to measure progress in the development of the inventory app.

This proposal marks a roadmap by which to further design and make the inventory app. The next step, of course, is to devise more technical designs regarding the interface, implement them as software, and then test and further refine them until a working inventory item management application is ready for production release.

References

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